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Case 34-2021: A 38-Year-Old Man with Altered Mental Status and New Onset of Seizures

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PRESENTATION OF CASE

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Dr. Luke A. Stevens (Emergency Medicine): A 38-year-old man was evaluated at this hospital because of altered mental status and a seizure.

The patient had been in his usual state of health until the night before the current evaluation. His wife reported that he fell out of bed at approximately 4 a.m. and was on the floor “shaking.” He appeared confused and was “speaking gibberish.” Police were called to the patient’s apartment, and emergency medical services were activated. On evaluation at the patient’s home, a fingerstick blood glucose measurement was 110 mg per deciliter (6.1 mmol per liter). The patient was combative and disoriented, and he actively resisted being placed in the ambulance. On arrival at the emergency department, he had a witnessed generalized tonic-clonic seizure, lasting 2 minutes, for which lorazepam was administered intravenously.

A limited history was obtained from the patient’s wife, brother, and sister-in-law. The patient had not been ill recently and had no history of seizures or cardiovascular, respiratory, gastrointestinal, genitourinary, or neurologic disorders. His medical history was notable for a laparoscopic appendectomy. The patient took no medications and had no known adverse reactions to medications. He worked in environmental maintenance at a local business. He lived with his wife, daughter, and son. He had immigrated to Boston from a rural area of Guatemala approximately 20 years earlier. He rarely drank alcohol and did not use tobacco or illicit drugs. There was no known family history of seizure disorder or other neurologic disease.

The temperature was 36.4°C, the heart rate 120 beats per minute, the blood pressure 171/90 mm Hg, the respiratory rate 22 breaths per minute, and the oxygen saturation 95% while the patient was using a nonrebreather mask. His eyes were open, and an involuntary upward gaze was noted; the pupils were 4 mm, symmetric, and reactive to light. He did not verbally respond to questions or follow commands. Gag and cough reflexes were normal. He withdrew his arms and legs in response to pain, and a jerking motion of the head was noted. The score on the Glasgow Coma Scale was 6 (on a scale of 3 to 15, with lower scores indicating greater alteration of consciousness). The neck was supple. Peripheral reflexes were

normal. The toes were downgoing bilaterally. Scant blood was observed in the mouth, without visible lacerations. The remainder of the examination was normal.

Blood levels of albumin, globulin, calcium, phosphorus, magnesium, lipase, N-terminal pro-B-type natriuretic peptide, and troponin T were normal, as were the prothrombin time, partial-thromboplastin time, and results of liver-function tests; other laboratory test results are shown in Table 1. A blood specimen was obtained to test for strongyloides antibodies. An electrocardiogram showed sinus tachycardia at a rate of 114 beats per minute but was otherwise normal.

Seven minutes after the first dose, a second dose of lorazepam was administered intravenously for suspected continued seizure activity. The patient remained confused and agitated, and an endotracheal tube was placed for airway protection. A chest radiograph was normal, with the endotracheal tube in an appropriate position.

A diagnosis was made.

EMERGENCY MANAGEMENT IN A PATIENT WITH SEIZURES

Dr. Jonathan E. Slutzman: In a patient presenting to the emergency department with possible seizure activity, management proceeds along two concurrent tracks: diagnostic and therapeutic. The therapeutic track involves both supportive and curative (frequently empirical) approaches. After airway management is established, the next step is to control the seizure to prevent further neurologic deterioration. First-line therapy is typically a parenteral benzodiazepine, either intramuscular midazolam or intravenous lorazepam (if intravenous access is readily available).¹ Preferred second-line agents include phenytoin or fosphenytoin, levetiracetam, and valproate, with no clear differences between the options in clinical effectiveness.² For patients in whom an acute infectious process, such as meningitis, is suspected, early empirical administration of antibiotic agents is indicated before lumbar puncture (if the procedure would delay the administration of antibiotics). Intracranial imaging is necessary in patients who have persistent abnormal mental status and is also recommended in adult patients with a first seizure who return to normal mental status, to identify additional primary causes, such as hemorrhage, neoplasm, or other mass lesions.³

DIFFERENTIAL DIAGNOSIS

Dr. Andrew J. Cole: This previously healthy 38-year-old man presented to the emergency department with confusion and possible seizure activity. I will discuss my general approach to evaluating a patient with an apparent first seizure, and then I will develop a differential diagnosis for this patient.

EVALUATION OF FIRST SEIZURE

From the neurologist's perspective, the first question to ask when evaluating a patient referred for an apparent first seizure is always, "Was the event a seizure?" In this case, the initial event, in which the wife found the patient confused, does not help to determine whether he had a seizure. The second event was witnessed by medical professionals and described as a generalized convulsion; the patient also had blood in his mouth, presumably from biting his tongue.

The second question is, "What is the cause of the seizure activity?" Identifying a cause not only helps the patient to understand the illness but also helps the neurologist to estimate the likelihood of recurrence. The cause ties directly into the third question, "What is the most appropriate treatment?" The neurologist must decide whether antiseizure treatment is required and whether there is a need to initiate long-term therapy.

Patient Assessment

Among patients presenting with an apparent first seizure, a detailed assessment results in confirmation of a first seizure in one third of cases. In another one third of cases, diagnostic testing confirms seizure activity but also identifies evidence of a previous seizure. Finally, in the remaining one third of cases, no seizure activity is identified.

For all these patients, obtaining the clinical history is key. The most powerful tool in the evaluation of a possible seizure is additional information. Speaking to a witness may reveal critical details that define the chronology of the disease process and indicate whether the seizure was provoked, which is the case in approximately 70% of patients with a first seizure. Emergency medicine providers are in a great position to interview family members and friends in order to find fleeting but critical details, such as

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Initial Evaluation, This Hospital
Blood		
Hemoglobin (g/dl)	13.5–17.5	15.0
Hematocrit (%)	41.0–53.0	45.9
White-cell count (per μ l)	4500–11,000	15,420
Differential count (per μ l)		
Neutrophils	1800–7700	8900
Lymphocytes	1000–4800	4960
Monocytes	200–1200	1330
Eosinophils	0–900	40
Basophils	0–300	50
Immature granulocytes	0–100	140
Platelet count (per μ l)	150,000–400,000	293,000
Sodium (mmol/liter)	135–145	142
Potassium (mmol/liter)	3.4–5.0	3.3
Chloride (mmol/liter)	98–108	98
Carbon dioxide (mmol/liter)	23–32	16
Urea nitrogen (mg/dl)	8–25	21
Creatinine (mg/dl)	0.60–1.50	1.16
Glucose (mg/dl)	70–110	147
Anion gap (mmol/liter)	3–17	28
Lactic acid (mmol/liter)	0.5–2.2	14.8
Vitamin B ₁₂ (pg/ml)	>231	559
Thyrotropin (μ U/ml)	0.40–5.00	1.29
HIV antibody and p24 antigen	Negative	Negative
Treponemal antibody	Nonreactive	Nonreactive
Acetaminophen (μ g/ml)	0.0–25.0	<5.0
Salicylates (mg/dl)	0.0–20.0	<0.3
Ethanol (mg/dl)	Negative	Negative
Arterial blood gases		
Fraction of inspired oxygen	—	0.21 (endotracheal tube)
pH	7.35–7.45	7.35
Partial pressure of carbon dioxide (mm Hg)	35–42	41
Partial pressure of oxygen (mm Hg)	80–100	73
Urine		
Color	Yellow	Yellow
Clarity	Clear	Clear
pH	5.0–9.0	5.0
Specific gravity	1.001–1.035	1.018
Glucose	Negative	Negative
Ketones	Negative	Negative
Leukocyte esterase	Negative	Negative

Table 1. (Continued.)

Variable	Reference Range, Adults†	On Initial Evaluation, This Hospital
Nitrite	Negative	Negative
Blood	Negative	+1
Protein	Negative	+1
Erythrocytes (per high-power field)	0–2	0–2
Leukocytes (per high-power field)	<10	<10
Bacteria	None	+1
Toxicology	Negative for amphetamines, barbiturates, benzodiazepines, cannabinoids, cocaine, opiates, and phen-cyclidine	Negative for amphetamines, barbiturates, benzodiazepines, cannabinoids, cocaine, opiates, and phen-cyclidine

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110. To convert the values for vitamin B₁₂ to picomoles per liter, multiply by 0.7378. To convert the values for salicylates to millimoles per liter, multiply by 0.07240. To convert the values for ethanol to millimoles per liter, multiply by 0.2171.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

whether there was a change in the patient's behavior, health, sleep, alcohol or substance use, or routines that could be a provoking factor; whether there were any warning signs; and whether the patient's seizure was focal at the onset. The history may also reveal whether the first seizure was missed. A typical scenario is that the early seizures manifest as partial seizure events with alterations of consciousness and unusual behavior, and it is the generalized convulsion that brings the patient to medical attention and illuminates the diagnosis.

For this patient, a history was obtained from family members. On the day before presentation to the emergency department, he had been taking care of his children and had eaten dinner with his brother; there was no report of unusual or altered behavior at dinner. Furthermore, there was no history of recent sleep deprivation. With regard to potential medications that can provoke seizure, we ask whether the patient had discontinued long-term benzodiazepine therapy or taken proconvulsant medications, such as tramadol or bupropion. This patient had been taking no prescribed or over-the-counter medications.

A neurologic examination can help to assess for brain dysfunction, whether focal or generalized, and to identify any previously unrecognized

conditions. This patient had no physical findings that were suggestive of an underlying disease.

Laboratory Testing

In addition to history taking and neurologic examination, targeted laboratory testing, cerebral imaging, and electroencephalography (EEG) can inform the diagnostic evaluation of a potential first seizure. In this patient, the laboratory evaluation ruled out hyponatremia, renal dysfunction, and liver dysfunction. A complete blood count with differential count can provide evidence of infection, although leukocytosis associated with demargination (which was seen in this patient) is not uncommon in the postictal period. The patient's urine and serum toxicology panels were negative. An elevated lactic acid level is consistent with prolonged muscle activity characteristic of a tonic-clonic seizure, and it is more suggestive of seizure than of a condition that mimics seizure. Cerebrospinal fluid analysis is reserved for specific situations that are highly suggestive of a viral or bacterial infection involving the central nervous system (CNS).

Cerebral Imaging

Magnetic resonance imaging (MRI) and computed tomography (CT) of the head can be used

to evaluate patients with seizures. In general, MRI is the preferred technique, because it has higher sensitivity and specificity than CT for the detection of brain parenchymal lesions.⁴ In the emergency department, specific MRI protocols for seizure evaluation, stratified according to age group, can be used to determine the most likely associated abnormalities. Nonetheless, CT is commonly used in the emergency department, because it can rule out acute hemorrhage, masses, and calcified lesions more rapidly than MRI. It is likely that this patient underwent CT after arrival and stabilization in the emergency department, but if the CT study was negative, MRI would ultimately help us to carefully assess for a causative anatomical abnormality.

Electroencephalography

EEG is extremely useful in classifying the seizure problem, characterizing the seizure as focal or generalized, and identifying the location in the brain from which it arises.⁴ An interictal EEG is abnormal in up to 60% of patients with epilepsy, and sleep deprivation and sleep recordings increase the yield of informative results. EEG can also be used to identify whether the patient is still seizing (if consciousness is slow to be regained) and to assess the risk of recurrence. Findings on EEG may help physicians to determine whether some patients need benzodiazepine, sedation, and airway protection. In addition, the findings may help physicians to appropriately adjust the dose of benzodiazepine so that oversedation does not occur. More than 90% of seizures end spontaneously in 2 to 3 minutes.⁵ It would be informative to review an EEG obtained immediately after the patient received the first dose of lorazepam, to assess whether epileptic activity was suppressed and whether there was ongoing seizure activity. However, EEG is rarely, if ever, performed rapidly in the emergency department.

DIFFERENTIAL DIAGNOSIS OF FIRST SEIZURE IN THIS PATIENT

There is no evidence of a provoked event in this patient's history or examination; he had appeared well the day before the seizure. In addition, there is no evidence of a previous neurologic, medical, or psychiatric condition that would have put him at increased risk for seizures. It appears that this 38-year-old man had been healthy.

However, he had emigrated from a rural area of Guatemala, and therefore, we need to consider endemic infectious diseases that could have increased this patient's risk for seizures even years after exposure, such as a parasitic brain infection.

Cysticercosis

Cysticercosis is the most common cause of acquired epilepsy worldwide. The disease results from ingestion of eggs from the tapeworm *Taenia solium*. At first, the disease is relatively indolent, because the eggs form cysts that do not generate a clinically significant immune response for approximately 5 years. In the 1930s, MacArthur and Dixon observed the disease process in former soldiers who had returned to England after serving in India.^{6,7} They noted that seizures associated with cysticercosis often emerged years after the initial exposure, alongside an inflammatory response associated with late calcification of the parasitic lesion. This disease is endemic in areas of Asia and Central America⁸; in these areas, 10 to 50% of patients with epilepsy have evidence of neurocysticercosis on cerebral imaging. There are also locations in the United States where cysticercosis is highly prevalent, including Southern California, Texas, and New York City.

Toxoplasmosis

Toxoplasmosis is another parasitic disease that can cause seizures in a person who appears to be otherwise healthy. The disease is distributed worldwide, with a prevalence of up to 80% in certain Central American and South American countries. Patients with toxoplasmic encephalitis present with fever, headache, confusion, and seizures; imaging studies may reveal multiple ring-enhancing lesions with edema. However, such manifestations are rare in immunocompetent persons. To our knowledge, this patient had no risk factors for human immunodeficiency virus (HIV). Furthermore, he had a negative HIV screening test and had no evidence of leukopenia, lymphopenia, or hepatic or renal dysfunction; these findings make reactivation of toxoplasmosis unlikely.

Cancer

Cancer is another possible cause of a first seizure in adults. In particular, patients with low-grade brain tumors frequently present with focal

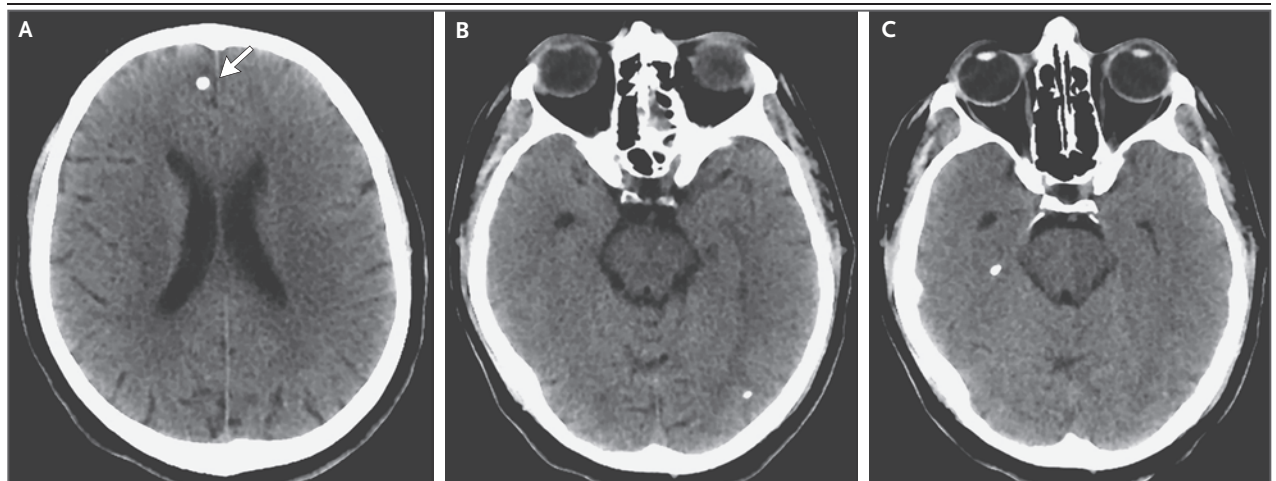


Figure 1. CT of the Head.

CT of the head was performed without the administration of contrast material. Axial images show coarse intraparenchymal calcifications in the anteromedial right frontal lobe (Panel A, arrow), the left occipital lobe (Panel B), and the right medial temporal lobe (Panel C). The right frontal calcification is associated with mild surrounding hypodensity.

seizures in the absence of focal signs and symptoms preceding the event. In this case, imaging studies would be needed to assess for and rule out a tumor, because there is no evidence in the history or examination results that would fully rule out this possibility.

A cryptogenic, genetic, or idiopathic generalized epilepsy syndrome should be considered. Idiopathic generalized epilepsy syndrome usually occurs with a stressor such as fever or sleep deprivation. These syndromes are not particularly likely in a 38-year-old person, but EEG would be informative.

Conditions That Mimic Seizures

We must always be mindful of conditions that mimic seizures and consider the possibility that the presenting event was not really a seizure. However, in this patient, a tonic-clonic seizure was witnessed, so the epileptic nature of the event seems clear. Cerebrovascular accidents and transient ischemic attacks sometimes mimic seizures, but this patient had no evidence of focal brain dysfunction and his event included positive signs rather than evidence of loss of function. Intoxications, syncope, migraine, behavioral dysregulation, and psychiatric events represent diagnostic challenges. Again, toxicology panels were negative in this patient. Further neuropsychiatric interviewing and examination would help to rule out the other possibilities.

On the basis of the features of the patient's presentation, the fact that he had been healthy the day before the seizure, and his history of living in a rural area of Guatemala, neurocysticercosis is the most likely diagnosis in this case. To establish this diagnosis, CT was most likely performed, followed by MRI and EEG.

DR. ANDREW J. COLE'S DIAGNOSIS

Neurocysticercosis.

IMAGING STUDIES

Dr. Michael H. Lev: In the emergency department, the patient underwent CT of the head, performed without the administration of intravenous contrast material. The CT study was notable for the presence of several coarse intraparenchymal calcifications, the largest of which was located in the anteromedial right frontal lobe (Fig. 1), with mild surrounding hypodensity that was suggestive of vasogenic edema. These findings are typical of neurocysticercosis. Other calcified lesions, such as chronic granulomatous disease and oligodendroglioma, can have a similar appearance but are unlikely in this case.

In the emergency department, the patient also underwent MRI of the head, performed after the administration of intravenous gadolinium, for further characterization of the findings noted

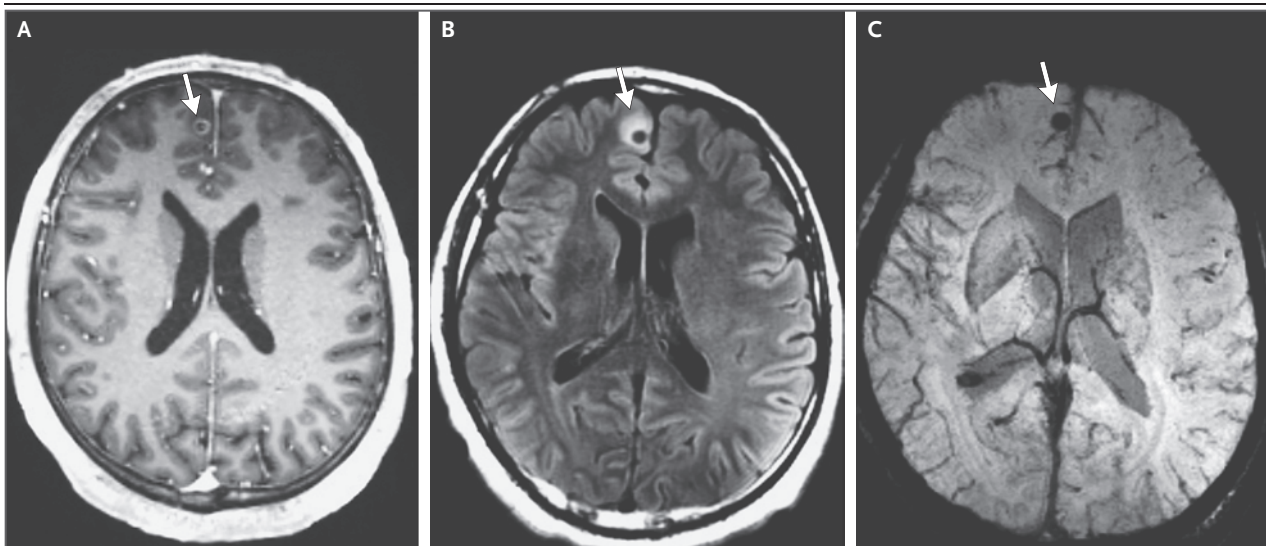


Figure 2. MRI of the Head.

MRI was performed after the administration of intravenous gadolinium. Axial images show a focal enhancing lesion in the anteromedial right frontal lobe. The right frontal lesion is associated with ring enhancement with a possible central punctate focus on a T1-weighted image (Panel A, arrow), mild surrounding hyperintensity on a fluid-attenuated inversion recovery image (Panel B, arrow), and susceptibility effect on a gradient-echo image (Panel C, arrow).

on CT. The MRI study showed focal enhancing lesions in the right frontal lobe, left occipital lobe, and right temporal lobe that corresponded to the coarse calcifications seen on CT. The right frontal lesion was associated with ring enhancement with a possible central punctate focus on T1-weighted imaging (Fig. 2A), mild surrounding hyperintensity on fluid-attenuated inversion recovery imaging (Fig. 2B), and susceptibility effect on gradient-echo imaging (Fig. 2C). This constellation of CT and MRI findings strongly suggests a diagnosis of neurocysticercosis. Other lesions such as those caused by tuberculosis, toxoplasmosis, and cancer are unlikely. Neurocysticercosis can be present in intraparenchymal, intraventricular, meningeal, spinal, and ocular locations. The forms can be active (viable cysts with scolex, rarely associated with symptomatic disease), transitional (degenerating cysts in the colloidal and necrotic stages with adjacent edema, commonly associated with seizures), or inactive (calcifications or meningeal fibrosis).⁹

PATHOLOGICAL DISCUSSION

Dr. George Eng: When imaging findings are suggestive of cysticercosis, the diagnosis can be con-

firmed with serologic studies. Direct detection of parasite antigens in patient specimens is not standard but may be considered when neuroimaging is not available.¹⁰ Conversely, testing for anti-cysticercal antibodies can be performed at reference laboratories with the use of either an enzyme-linked immunoelectrotransfer blot (EITB) assay or an enzyme-linked immunosorbent assay (ELISA). The EITB assay identifies anti-cysticercal antibodies to lentil-lectin affinity-purified cyst antigens and has higher sensitivity than the ELISA that uses crude antigens.¹¹ Furthermore, the specific banding pattern of immunoreactivity on the EITB assay can distinguish cysticercosis from other helminthic infections.^{12,13} Thus, the EITB assay is the recommended test for confirming the diagnosis of cysticercosis; however, its sensitivity depends on the number and stage of the cysts.^{14,15} The sensitivity can be low with only calcified lesions present, but it increases with the number of living cysts.¹¹

In this patient, an EITB assay performed to test for the presence of anti-cysticercal antibodies was negative. The patient had evidence of three brain lesions, with one showing ring enhancement on MRI and all three showing partial calcification on CT. The calcifying progression

of the lesions may account for the absence of antibody positivity in this patient. In addition, tests for antibodies against toxoplasma, strongyloides, and treponema were negative, as were an interferon-gamma release assay and a purified protein derivative tuberculin skin test for tuberculosis.

DISCUSSION OF MANAGEMENT

Dr. Edward T. Ryan: Consensus guidelines regarding the diagnosis and treatment of patients with neurocysticercosis have been published by the Infectious Diseases Society of America and the American Society of Tropical Medicine and Hygiene.¹⁴ The diagnosis usually relies on recognition of an appropriate clinical situation with supportive epidemiologic and neuroimaging features and less often relies on serologic testing. The consensus guidelines recommend that both CT and MRI, the latter with three-dimensional volumetric sequencing such as fast imaging employing steady-state acquisition (FIESTA) sequences,^{16,17} be performed for the evaluation of patients with probable neurocysticercosis. CT can delineate calcifications; MRI provides finer resolution, may allow for visualization of internal structures such as protoscolices, and assesses the degree of lesional enhancement and perilesional edema. The consensus guidelines suggest that the EITB assay is the preferred serologic test.

Four approaches should be considered in the treatment of patients with neurocysticercosis: use of antiparasitic agents such as albendazole and praziquantel, use of antiinflammatory agents such as glucocorticoids, use of antiseizure agents (if seizures are part of the clinical scenario), and mechanical interventions such as placement of a ventriculoperitoneal shunt or endoscopic removal of intraventricular cysts (if indicated). For each patient, the appropriate treatment is determined by the clinical situation and the type of parasitic involvement of the CNS.

Patients with viable intraparenchymal cysts are usually treated with antiparasitic agents and glucocorticoids, as well as antiseizure agents if seizures have occurred. The appropriate dosage and duration of glucocorticoid therapy are unclear. The administration of antiparasitic agents in this situation is associated with an increased likelihood that the lesions on imaging will resolve and a decreased risk of subsequent sei-

zures. The consensus guidelines recommend that patients with three or more viable intraparenchymal cysts be treated with combination antiparasitic therapy (albendazole and praziquantel) and that patients with one or two viable cysts be treated with albendazole alone. For patients with a sole cyst that is degenerating, treatment with albendazole, glucocorticoids, and antiseizure agents is recommended. For patients with isolated calcifications, the consensus guidelines recommend against treatment with an antiparasitic agent. Antiseizure medication is indicated in these patients if seizures are part of the clinical scenario. Perilesional edema and enhancement have been reported even with apparently dead and calcified neurocysticercosis lesions, perhaps resulting from leakage of retained antigens. Because there have been case reports of recurrence of edema and seizures after tapering or cessation of glucocorticoids, the consensus guidelines suggest that glucocorticoids should be used with caution, if at all, in patients with isolated pericalcification edema and enhancement. Patients with subarachnoid, ocular, spinal, or encephalitic cysticercosis also have specific treatment algorithms.

The consensus guidelines recommend that patients who will receive prolonged glucocorticoid therapy as part of their treatment regimen first be evaluated for latent tuberculosis and be either evaluated or empirically treated for intestinal strongyloidiasis. All patients with neurocysticercosis should undergo funduscopic evaluation before the initiation of antiparasitic agents. The choice of antiseizure medication can be based on local availability. The antiseizure therapy should be continued for at least 2 years, and cessation of therapy can be considered if the patient remains seizure-free and is deemed to be at low risk for recurrence.¹⁴ Guidelines also recommend that, if it is thought that the patient with neurocysticercosis had become infected in an area in which the disease is not endemic, members of the household and other close contacts should be evaluated for the presence of an adult tapeworm, and local public health authorities should be appropriately notified.

This patient had a negative test for latent tuberculosis, and he was empirically treated with ivermectin for possible concomitant strongyloidiasis. Results of the funduscopic examination were normal.

FOLLOW-UP

Dr. Stevens: The patient was admitted to the neurosciences intensive care unit, and the lactic acid level and white-cell count both normalized within hours. Levetiracetam therapy was started to control seizures. An EEG showed generalized polymorphic delta and theta slowing of the background and generalized rhythmic delta activity and fast discharges. The patient was extubated 12 hours after presentation and was transferred to the neurology service the following morning. He started treatment with 2 weeks of albendazole and praziquantel, along with 4 weeks of high-dose prednisone, followed by a 4-week tapering course. He was discharged on hospital day 5, with normal results on a neurologic examination and no further seizure activity. Repeat MRI of the head performed 4 months and 10 months after presentation revealed a decrease in edema around the right frontal lesion. Three years after the first seizure, the patient has remained seizure-free and continues to take levetiracetam.

A Physician: Dr. Ryan, would you comment on the treatment options for this patient?

Dr. Ryan: The imaging findings and time

course were most consistent with seizures related to nonviable calcified cysticerci with edema. The patient was treated with antiparasitic medications, but after reviewing this case, it is clear that primary treatment would target the seizures and not include antiparasitic agents.

A Physician: Dr. Cole, when would you recommend stopping the antiseizure medication in this patient?

Dr. Cole: The issue of when to stop the medication is problematic, because the calcified lesion will remain in perpetuity. Even a negative EEG obtained several years after this event would not be especially reassuring, although a positive EEG would be difficult to ignore. Overall, I make these decisions with the patient after a detailed discussion of both the risks and the benefits. I personally would be reluctant to stop the medication unless the patient insisted.

FINAL DIAGNOSIS

Seizure from neurocysticercosis.

This case was presented at Emergency Medicine Grand Rounds. Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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